

Turbulence modulation via dilute surfactant additives in experimental Taylor–Green flow

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Abstract here.

I. INTRODUCTION

Since the discovery that trace amounts of dissolved polymer could lead to drag reduction in pipe flows [1], the use of additives for drag reduction in wall bounded flows has become established. Besides polymer solutions, it has since been discovered that self-assembling surfactant solutions share similar effects [2], owing to the fact that dilute surfactant solutions can form wormlike micelles whose relaxation dynamics mimic that of polymer chains while resisting mechanical degradation. Beyond the applications of drag reduction, it has shown that viscoelastic additives are known to modulate turbulence more broadly, reshaping the energy spectrum and producing an elastic range governed by additive elasticity rather than inertial at the small scales [3, 4], with surfactant-specific micellar structures shown to alter vortex shedding and turbulence statistics directly [5].

In spite of academic and commercial interest, comparatively few studies address the behavior of dilute surfactant solutions in the context of three-dimensional bulk turbulence, with most of the literature focused on wall-bounded or quasi-two-dimensional flows. This is in part related to the fact that the rheology of very dilute surfactant solutions remains difficult to pin down, with reported values of viscosity and relaxation time of CTAC/NaSal solutions (such as the considered in the present experiment) varying considerably between studies [6, 7]. In spite of the diverging measurements and reports that some of the dilute solutions behave as a Newtonian fluid under rheological measurement [6], such solutions are known to produce measurable drag reduction [8], which further reinforces the need to analyze such solutions more in depth.

Motivated by this gap, here we present an experimental analysis of the turbulence generated by a four-rotor, Taylor–Green-like experiment using dilute CTAC solutions, in an attempt to characterize the modulation of turbulence caused by the addition of the surfactant across a wide range of Reynolds numbers for flows driven by inertial forcing.

II. EXPERIMENTAL DEVICE

In the present study, we employ an experimental device designed to produce Taylor–Green-like forced turbulence on a laboratory setting. The apparatus, with schematic shown on figure 1, consists of a water tank

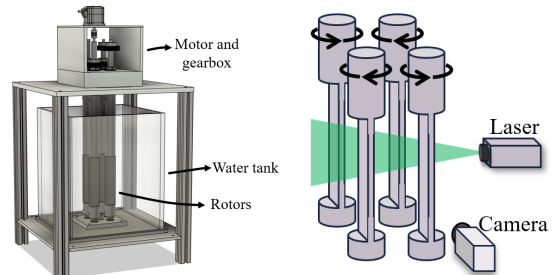


FIG. 1. CAD of the experimental device (left) and schematic showing normal orientation of camera and laser sheet, as well as direction of rotation of the rotors.

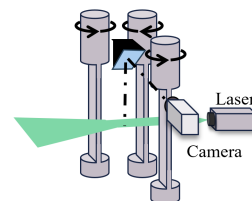


FIG. 2. Schematic of the altered orientation of camera and laser using mirror for imaging of the space between rotors. One of the rotors is omitted for clarity.

with four rotors which are driven by a gear system and rotate in opposing directions in order to produce turbulence in the region between them (hereinafter simply denominated bulk), which is the target of our studies. Spatially-resolved measurements are obtained via 2D particle tracking (briefly described in section II A) via a traditional laser-sheet and high-speed camera combination, which are positioned as also shown in figure 1 for the case of bulk imaging.

Aside from the imaging of the bulk region, it is also possible to change the relative orientations of the laser sheet and camera, and use a mirror to allow for top-down imaging of the region in the vicinity of the blades, as shown in the schematic of figure 2.

As shown in the schematics of figures 1 and 2, the rotors themselves are shaped as blades with rectangular cross-section. While the design of the experimental device allows for the geometry of the rotors to be changed between cylinders and the carved blades, the use of the latter allows us to isolate inertial forcing in the driving of the flow, and therefore ignore boundary-layer effects which would be present in the case of viscous driving which happens when cylindrical rotors are used. Addi-

tionally, the rotation direction is such that mean stretching of the flow happens in the out-of-plane component of the camera image (when imaging the bulk region in the vertical plane), and is accompanied by the formation of secondary vortices as shown in previous numerical studies [9]. For simplicity, the carved rotors will hereinafter simply be denominated blades.

While not shown in the schematics, the experimental device is also encased in thermal insulation and connected to an air-processor system. By controlling the temperature of the surrounding air, the water temperature was maintained at approximately 26 °C over the course of the experiments.

A. Data measurement and reduction

As previously mentioned, processing of the obtained images to calculate the velocity fields is conducted via a particle tracking velocimetry (PTV) code developed in-

house. Although the details of the algorithm are out of the scope of the current document, the code is built as a collection of previously established methods. For the detection of tracer particles from the images, we employ a variation of the Iterative Particle Reconstruction (IPR) method, which is commonly employed for volumetric particle tracking as part of the Shake-the-Box method (Jahn et al. 2021). The tracking of detected particle positions between the video frames is carried out by a variation of the Polar Coordinate-System Similarity (PCSS) method (Schindler, 2010).

B. Surfactant solutions

III. TURBULENCE CHARACTERIZATION

IV. RESULTS

V. CONCLUSIONS

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